

PERSONAL PROJECT PPT

VK DSA Sheet

Master Patterns. Solve Problems. Level Up.

A free, open-source, pattern-based DSA practice platform unifying LeetCode, GeeksforGeeks, TakeUForward and Codeforces — with built-in progress tracking, revision management, and a personal knowledge base.

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vk-dsa-sheet-aw7j.vercel.app



DSA prep is scattered across too many places

Every coding platform promotes its own problem set — but none of them give students one unified, pattern-based place to practice, track, and revise.

PLATFORM	WHAT THEY OFFER	WHAT'S MISSING
LeetCode	Huge question bank	No pattern roadmap, no cross-platform tracking
TakeUForward	Striver's SDE Sheet	Limited to their own curated list
GeeksforGeeks	Topic-wise problems	No unified progress tracking across platforms
Codeforces	Competitive problems	No structured DSA sheet

There is no completely free website combining LeetCode + GFG + TUF + Codeforces + Pattern Names + Progress Tracker in one place.

One place for everything a DSA learner needs



ALL Patterns

Not just topics like "Arrays" — real patterns like Sliding Window, Two Pointers, Kadane's Algorithm



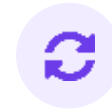
Multi-Platform

Questions pulled from LeetCode, GFG, TUF and Codeforces — all in one place



Completion Tracking

Mark every question done and watch progress bars fill up



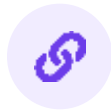
Revision Tracking

A separate track for what needs a second pass



Book Shelf

Store favourite DSA books alongside recommendations



Link Vault

Save YouTube videos, articles and editorials



Personal Notes

Write notes for any topic or question



Practice Calendar

A GitHub-style heatmap of daily coding consistency

"No such website existed. So I built one."

Pattern-based learning, not random practice

Instead of solving questions at random, learners work through the way interviewers actually think — by recognizing patterns.

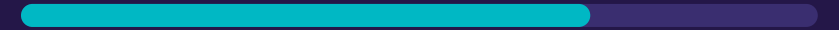
- 16 pattern categories: Arrays, Binary Search, Strings, Bit Manipulation, Linked List, Two Pointers, Sliding Window, Stacks & Queues, Greedy, Heaps, Binary Trees, BST, Graphs, Tries, DP and more
- Sub-pattern breakdown for each category (e.g. Arrays -> Rotate Array, Kadane's Algorithm, Dutch National Flag)
- Every pattern sources problems from LeetCode, TakeUForward, GeeksforGeeks and Codeforces

Arrays — Sub-patterns

Array Traversal 5/16



Rotate Array 5/7



Brute Force Subarray 1/9



Kadane's Algorithm 0/5



Dutch National Flag 0/4

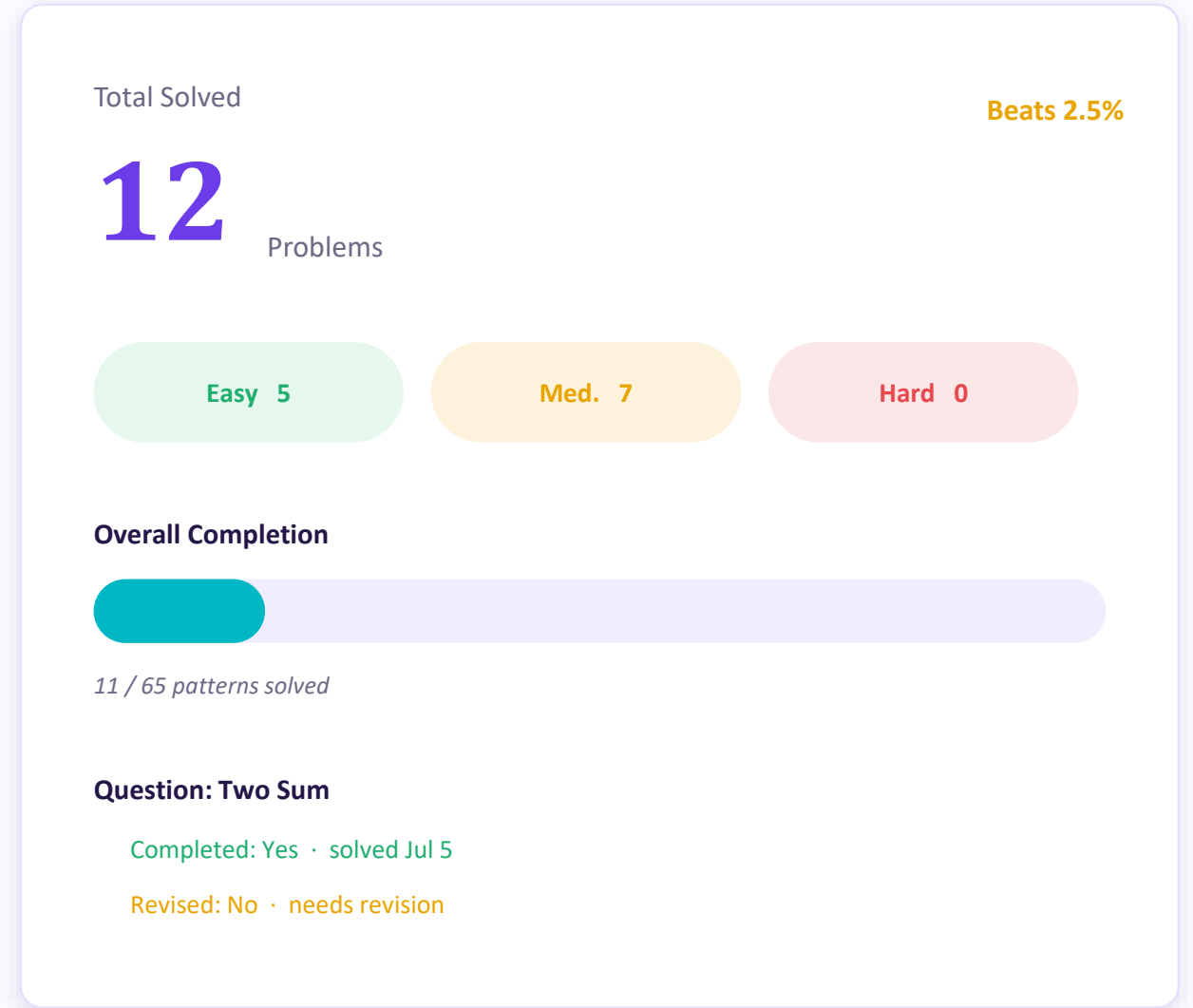


Platforms: LC · TUF · GFG · CF

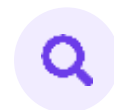
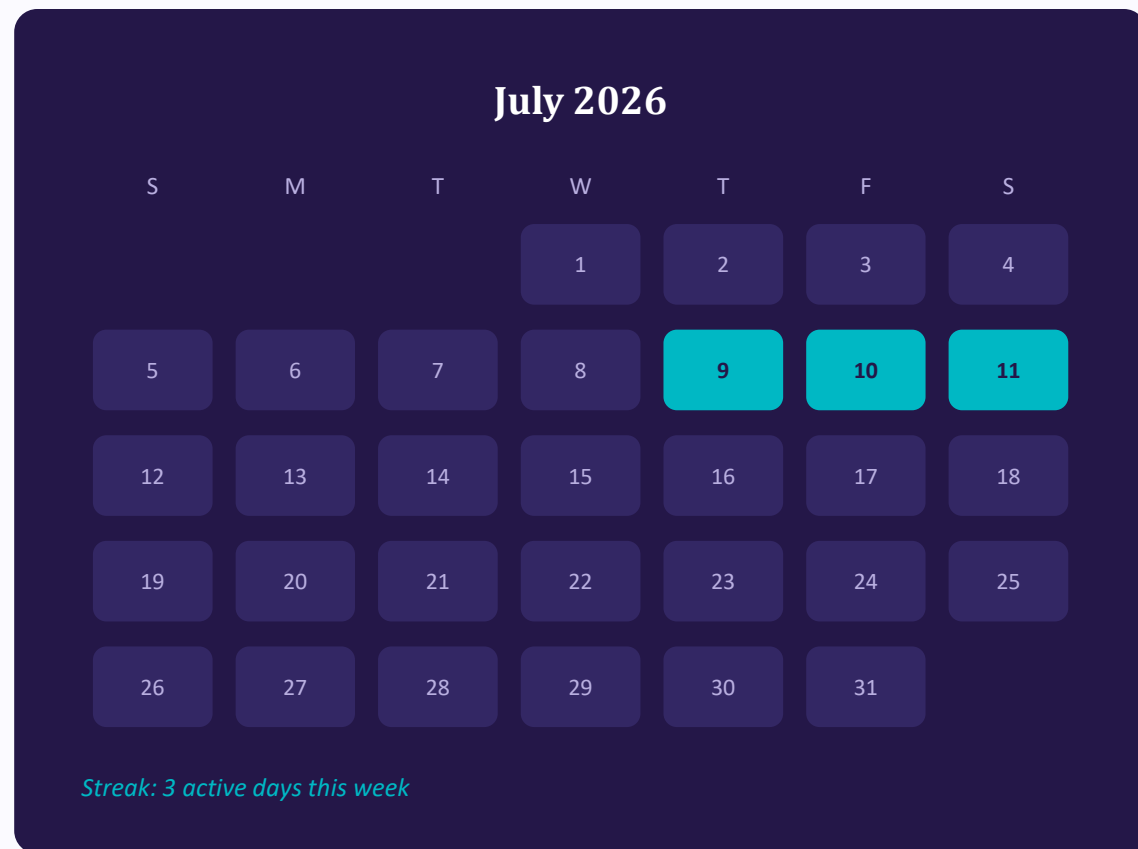
Two tracks: completion and revision

Every question can be marked complete AND separately flagged for revision — a feature missing from almost every existing sheet.

- 
Completion Tick
 Mark a question as done
- 
Revision Tick
 Separately flag for revisit
- 
Progress Bars
 Visual per sub-pattern, e.g. 5/16
- 
Beats %
 Compare against other learners



A practice calendar and a personal knowledge vault



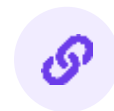
Search

Quickly find any pattern or question



Books

Recommended DSA books + your own favourites



Links

Save YouTube videos, articles, editorials



Notes

Personal notes for any topic or question

Secure, real-time and built to scale across devices



Firebase Authentication

Login / Sign up secures every account; progress, books, links and notes persist and sync in real time across devices.

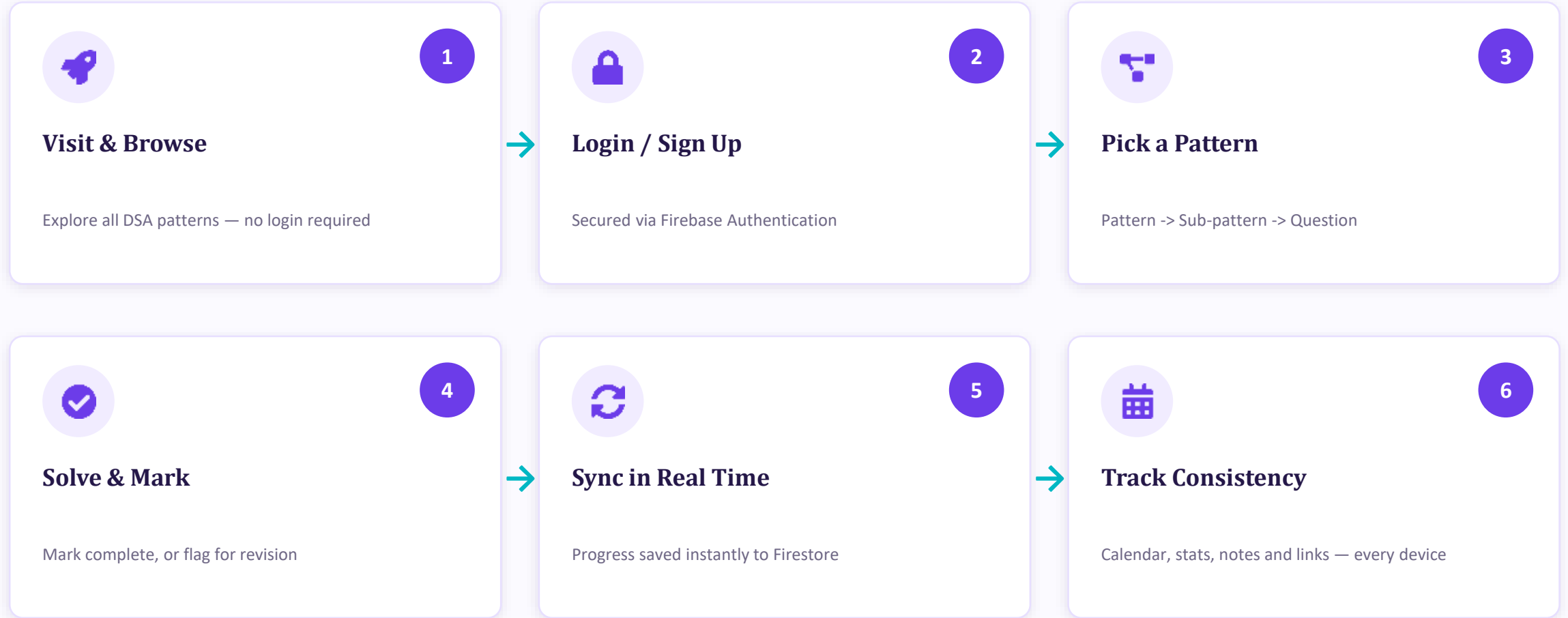
TECH STACK

Frontend	React · JavaScript · HTML5 · CSS3
Backend / Database	Firebase Authentication + Cloud Firestore
Deployment	Vercel
Version Control	Git & GitHub

Libraries & Tools

- **React** Component-based UI framework
- **Firebase Auth** User authentication (login / sign-up)
- **Cloud Firestore** Real-time database for user progress
- **React Router** Client-side routing and navigation
- **Custom CSS3** Dark theme styling with animation
- **Vercel** Production deployment and hosting
- **Git & GitHub** Version control and collaboration

From first visit to daily habit



16 pattern categories, one roadmap

01

Arrays

Traversal, Rotate, Kadane's, Dutch Flag

02

Binary Search

Standard, on Answers, on Rotated Arrays

03

Strings

Matching, Palindromes, Anagrams

04

Bit Manipulation

XOR Tricks, Bit Masking, Power of Two

05

Linked List

Reversal, Fast & Slow, Merge

06

Two Pointers

Opposite / Same Direction, Three Pointers

07

Sliding Window

Fixed, Variable, Shrinkable

08

Stacks & Queues

Monotonic Stack, Next Greater Element

09

Greedy Algorithms

Activity Selection, Fractional Knapsack

10

Heaps

Top-K, Merge K Sorted, Median

11

Binary Trees

Traversals, Views, Diameter, LCA

12

BST

Search, Insert, Delete, Validate

13

Graphs

BFS, DFS, Dijkstra, Topological Sort

14

Tries

Insert, Search, Prefix Matching

15

Dynamic Programming

1D/2D DP, Knapsack, LIS, LCS

16

More...

Additional patterns added regularly

How VK DSA Sheet compares

Feature	LeetCode	TUF Sheet	GFG Sheet	NeetCode	VK DSA Sheet
Pattern-wise Roadmap	—	~	—	~	✓
Multi-Platform Questions	—	—	—	—	✓
Progress Tracking	~	✓	—	✓	✓
Revision Tracking	—	—	—	—	✓
Practice Calendar	✓	—	—	—	✓
Personal Notes / Books / Links	—	—	—	—	✓
Completely Free	—	✓	✓	—	✓

The Dashboard, Live

Pattern categories, live progress ring, difficulty breakdown and a GitHub-style calendar — all visible the moment a learner logs in.



Built to feel premium, fast and personal



Dark Theme

Easy on the eyes for late-night coding sessions



Purple & Cyan

A modern, developer-friendly gradient identity



Responsive

Works cleanly across desktop, tablet and mobile



Real-Time

Firebase keeps every device instantly in sync

ROADMAP AHEAD

- Expand beyond 16 patterns with community-suggested additions
- Add contest/company-wise question tags (FAANG, product-based)
- Introduce streak rewards and leaderboard for friendly competition
- Mobile app version for on-the-go revision

Thank You

Built to give every DSA learner one home for practice, progress and revision.



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